

[← Go Back](#)

Hardware

GIGA Display Shield



Tutorials

[Getting Started with the GIGA Display Shield](#)[GIGA Display Shield GFX Guide](#)[GIGA Display Shield LVGL Guide](#)[GIGA Display Shield Draw Images Guide](#)[GIGA Display Shield Touch Interface](#)[GIGA Display Shield Automatic Screen Orientation with IMU](#)

GIGA Display Shield Microphone Guide

[GIGA Display Shield Camera Connector Guide](#)[GIGA Display Shield 3D Engine Tutorial](#)[SquareLine Studio With the GIGA Display Shield](#)[Display Camera Feed on GIGA Display Shield with OpenMV](#)[GIGA Display Shield emWin Guide](#)[GIGA Display Shield AppWizard Guide](#)[GIGA Display Shield Display Image From USB](#)

Home / Hardware / GIGA Display Shield / [GIGA Display Shield Microphone Guide](#)

GIGA Display Shield Microphone Guide

Learn how to use the GIGA Display Shield's Microphone

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The GIGA Display Shield has an embedded MEMS microphone (**MP34DT06JTR**) that when combined with the visual element of the GIGA Display Screen can be used in a number of ways.

Stream microphone data (plot it / print it),

Display volume as a bar,

Detect a clap or other noises

Using the [Arduino_Graphics](#), [Arduino_GigaDisplay_GFX](#) and [lvgl](#) we can also create animations and screen changes based on microphone data.

In this guide we will take a closer look at the [PDM library](#), and provide some examples that can be used with this shield.

Hardware & Software Needed

[Arduino GIGA R1 WiFi](#)[Arduino GIGA Display Shield](#)[Arduino IDE](#)

ON THIS PAGE

Hardware & Software Needed

[Downloading the Library and Board Package](#)[PDM Library](#)[Microphone PDM Example](#)[Microphone Examples](#) +[Next Step](#)[Help](#)

Downloading the Library and Board Package

Make sure the latest GIGA Board Package is installed in the Arduino IDE. You can install it directly in the IDE by navigating to the board manager and searching for **Arduino Mbed OS Giga Boards** and install it.

The **PDM library** is included in the Board Package, as well as the video driver library, **Arduino_H7_Video library**. Some examples in this guide uses other libraries that are listed in each example.

PDM Library

Pulse Density Modulation (PDM) is a technique used to convert analog signals into a digital 1-bit stream.

The **PDM library** is library built-in to the GIGA Board Package and allows you to read and process PDM signals. In this case, it reads the signal from the

Source code is available [here](#)

Library documentation is available [here](#)

Microphone PDM Example

The GIGA R1 Board Package includes a sample sketch called **PDM**, and in this section it is explained in more detail.

First we need to define the number of output channels, output frequency, a variable for counting when reading from the buffer and creating the buffer which the readings will be put into. This is done with the following lines:

```
1 // default number of out
2 static const char channe
3
4 // default PCM output fr
5 static const int frequen
6
7 // Buffer to read sample
8 short sampleBuffer[512];
9
10 // Number of audio sampl
11 volatile int samplesRead
```

A callback function needs to be set, which is called when new PDM data is ready to be read. We do this in the `setup()` function using:

```
1 PDM.onReceive(onPDMdata);
```

`onPDMdata` is the callback function that we will have to create at the end of the sketch.

Now when we want to print or use the readings let's do it with a `for` loop since they are inside a buffer, which we need to step through. But let's first check so that there are readings to be printed with a simple `if` statement. These lines will step through the buffer until all the readings inside are printed and then start over:

```
1  if (samplesRead) {  
2      for (int i = 0; i < s  
3          Serial.println(samp  
4      }  
5  }
```

Its inside this `for` loop where we can get readings that will then change visual elements on the screen. This is where you would put any code that needs to react to the microphone readings.

And the last important part is the callback function that we used in the `setup()` function. This will take care of reading the values into the buffer and setting the value of the `samplesRead` variable which we used in the previous step.

```
1  /**  
2   * Callback function to  
3   * NOTE: This callback i  
4   * Therefore using `Seri  
5   * */  
6  void onPDMdata() {  
7      // Query the number of  
8      int bytesAvailable = P  
9  
10     // Read into the sampl  
11     PDM.read(sampleBuffer,  
12  
13     // 16-bit, 2 bytes per  
14     samplesRead = bytesAva  
15 }
```

The full example is available in the [PDM Example](#) section just below.

Microphone Examples

In this section you will find a series of examples that uses the microphone.

PDM Example

This sketch can be found in **File > Examples > PDM** in the Arduino IDE. It reads the microphone data, stores it in a buffer and prints it to the Serial Monitor / Serial Plotter tool in the IDE.

```
1  #include <PDM.h>
2
3  // default number of ou
4  static const char chann
5
6  // default PCM output f
7  static const int freque
8
9  // Buffer to read sampl
10 short sampleBuffer[512]
11
12 // Number of audio samp
13 volatile int samplesRea
14
15 void setup() {
16   Serial.begin(9600);
17   while (!Serial);
18
19   // Configure the data
20   PDM.onReceive(onPDMda
21
22   // Optionally set the
23   // Defaults to 20 on
24   // PDM.setGain(30);
25
26   // Initialize PDM wit
27   // - one channel (mon
28   // - a 16 kHz sample
```

Open the Serial Monitor / Serial Plotter to see the data in real time.



Example sketch printing
values in the serial
monitor

Clap Detection Sketch (ArduinoGraphics)

This sketch uses the [Arduino_Graphics library](#) to change the color of the background when a loud noise is detected, such as a clap.

```
1 #include "Arduino_H7_Vi
2 #include "ArduinoGraphi
3 #include <PDM.h>
4
5 Arduino_H7_Video Displa
6
7 // default number of ou
8 static const char chann
9
10 // default PCM output f
11 static const int freque
12
13 // Buffer to read sampl
14 short sampleBuffer[512]
15
16 // Number of audio samp
17 volatile int samplesRea
18
19 void setup() {
20     delay(3000);
21     Display.begin();
22     Display.beginDraw();
23     Display.background(25
24     Display.clear();
25
26     PDM.onReceive(onPDMda
27
28     if (!PDM.begin(channe
```

After uploading a sketch, snap your fingers or clap your hands to produce a noise that meets the threshold. You should see the

Volume Indication Sketch (LVGL)

This sketch requires the [lvgl library](#), please make sure that is installed before you upload the sketch. The sketch will show a bar on the screen that is animated when noise is made, functionally making it display the volume of the microphones readings. You will find the full sketch just below.

 For more information about using LVGL with the GIGA Display Shield, take a look at our documentation [here](#).

```
1  #include <PDM.h>
2  #include "Arduino_H7_Vi
3  #include "lvgl.h"
4
5  Arduino_H7_Video
6
7  static void set_slider_
8      lv_bar_set_value((lv_
9  }
10
11 // default number of ou
12 static const char chann
13
14 // default PCM output f
15 static const int freque
16
17 // Buffer to read sampl
18 short sampleBuffer[512]
19
20 // Number of audio samp
21 volatile int samplesRea
22
23 lv_obj_t * obj;
24 lv_anim_t a;
25 int micValue;
26
27 void setup() {
28     delay(3000);
```

Next Step

Now that you know how to use the on-board microphone, feel free to explore the shield's other features, like the IMU with our [Orientation tutorial](#). Or if you rather dive deeper into LVGL, take a look at our [LVGL guide](#).

For the complete documentation for this shield, check out the [GIGA Display Shield](#) documentation page.

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